

Big O Notation

Time complexity of an algorithm

"How much time it takes to run a function as the size of the input grows."

Runtime

array
number of elements
 $n=5$

Const

array1 = [, , , , ]

number of elements
 $n=5$

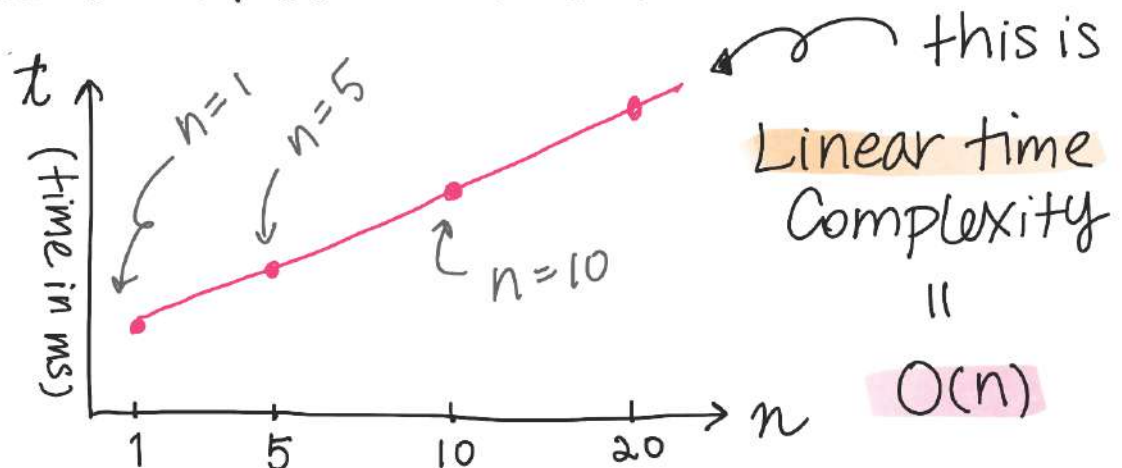
Let's see if there is a needle in the haystack!

```
JS Const numNeedles = (haystack, needle) => {  
  let count = 0  
  for (let i = 0; haystack.length; i++) {  
    if (haystack[i] === needle) count += 1;  
  }  
  return count;  
}
```



How long does it take to execute when the number of elements (n) is:

execution time grows linearly as array size increases!



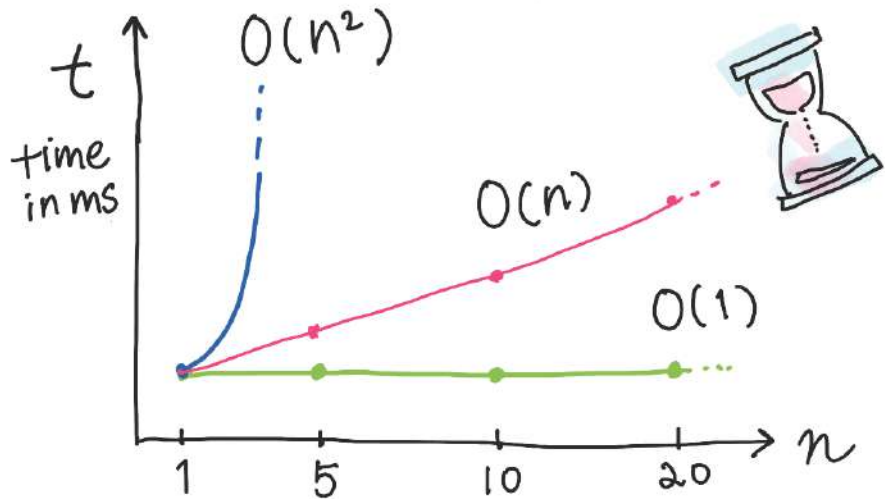
Big O Notation

JS Let's see if we have some function that doesn't actually loop the array:

```
const alwaysTrueNoMatterWhat = (haystack) => {  
  return true;  
}
```

$n=5$
 $n=10$
 $n=20$
⋮
↖ Array size has no effect on the runtime

☆ Constant time
||
 $O(1)$



☆ Quadratic time = $O(n^2)$

Const

array2 = [🐱, 🌸, 🍰, 🍰, ✂];

$n=5$, however the runtime is proportional to n^2

JS Const hasDuplicates = (arr) => {
 for (let i = 0; i < arr.length; i++)
 let item = arr[i];
 if (arr.slice(i+1).indexOf(item) !== -1) {
 return true;
 }
 return false;
}

① Loop thru the array

② Another array lookup w/ indexOf method

Data Structures

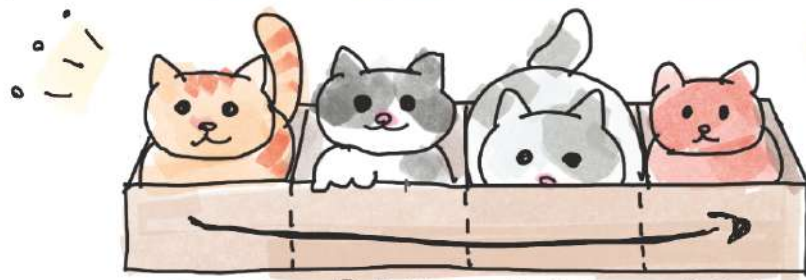
Array & Linked List

Array a linear data structure, stored in contiguous memory locations.



memory

Address 200 204 208 212



Array [0] [1] [2] [3]

$a[0]$
 $a[1]$
 $a[2]$
 $a[3]$

196
200
204
208
212
216
220
224
228

- ♥ Assume each 🐱 is an integer = requires 4 bytes space
- ♥ The array of 🐱 must be allocated contiguously!
→ address 200 — 216



🎉 yay!

- ♥ can randomly access w/ index
 $a[2] \rightarrow$ 🐱
- ♥ contiguous = no extra memory allocated = no memory overflow

👎 meh!

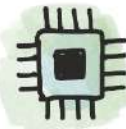
- 💀 fixed size. Large space may not be avail for big array
= 🐱 took the space! =
- 💀 Insert & delete elements are costly.
→ may need to create a new copy of the array & allocate at a new address.

Linked list

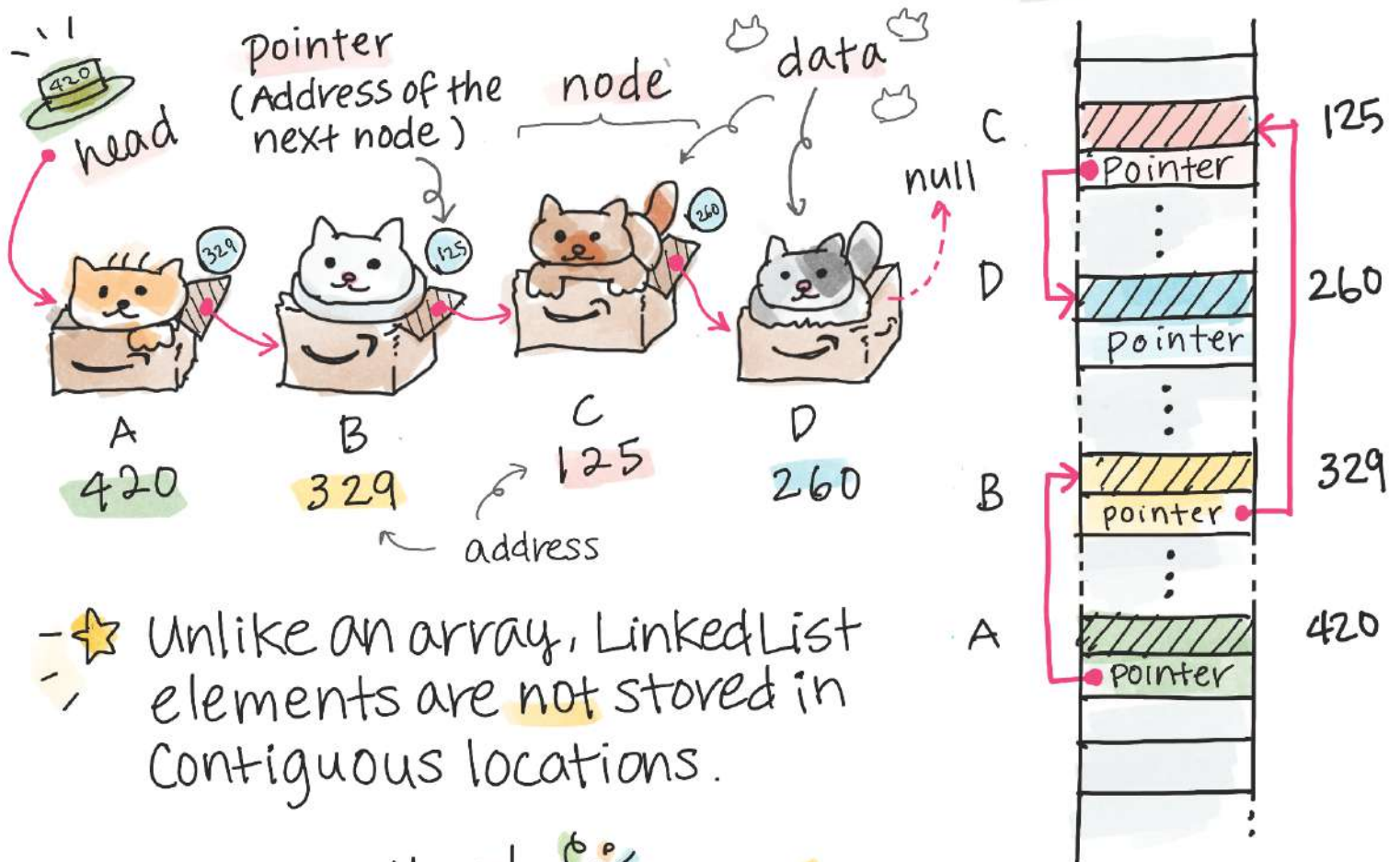
Array & Linked List

Data Structures

- ★ a linear data structure
- ★ each element is a separated object & elements are linked w/ pointers



memory



- ★ Unlike an array, LinkedList elements are not stored in Contiguous locations.

Yay! 🎉

- ♥ Dynamic data
= size can grow or shrink
- ♥ Insert & delete element are flexible.
→ no need to shift nodes like array insertion
- ♥ memory is allocated at runtime

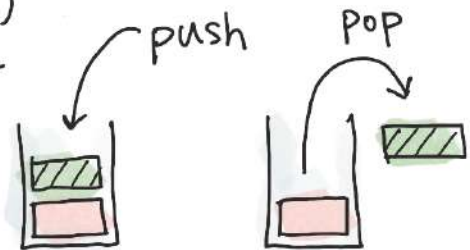
👎 meh!

- ☹ No random access memory.
→ Need to traverse n times
→ time complexity is $O(n)$. array is $O(1)$
- ☹ Reverse traverse is hard

Data Structures

Stack & Queue

A stack is a LIFO (Last-in-First-out) data structure, where an element added last (=push) gets removed first (=pop)



♡ just like a stack of ice cream scoops!



☆ Stack as an array in JS

arrays in JavaScript are dynamic!

```
let stack = [ ];  
stack.push('mint choc');  
stack.push('vanilla');  
stack.push('strawberry');  
let eaten = stack.pop();
```

stack is:
['mint choc', 'vanilla', 'strawberry']
eaten is 'strawberry'
['mint choc', 'vanilla']

♡ Time complexity is $O(1)$ for both pop + push.

omg, the bottom one is always melting!!

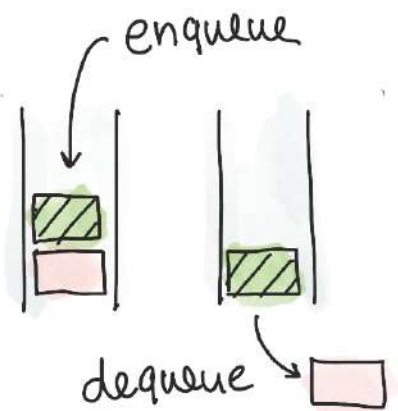
Data Structures

Stack & Queue

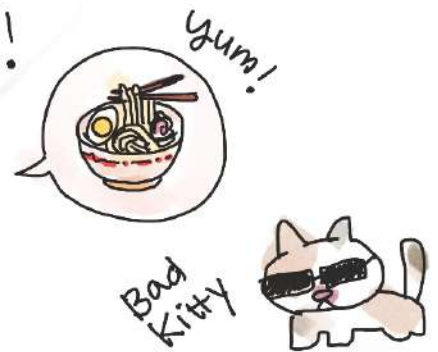
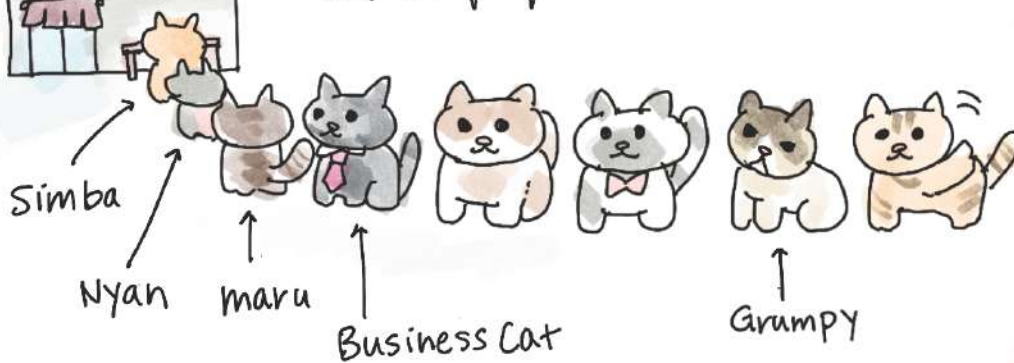
LIFO

FIFO

A queue is a FIFO (First-in-First-out) data structure, where an element added first (= enqueue) gets removed first (= dequeue)



just like waiting in line at a popular restaurant!



☆ Stack as an array in JS

```
let queue = [ ];  
queue.push('Simba'); // ['Simba']  
queue.push('Nyan'); // ['Simba', 'nyan']  
queue.push('maru'); // ['Simba', 'nyan', 'maru']  
let eater = queue.shift(); // eater is 'Simba'
```

Time Complexity should be $O(1)$ for both enqueue + dequeue but JS `shift()` is slower!

if you queue.unshift('bad Kitty'), instead of push(), then the cat cuts in to the front of line!

Wrong!

Data Structures Hash Table

- ♥ A hash table is used to index large amount of data
- ♥ Quick key-value look up. $O(1)$ on average
 - ↳ Faster than brute-force linear search

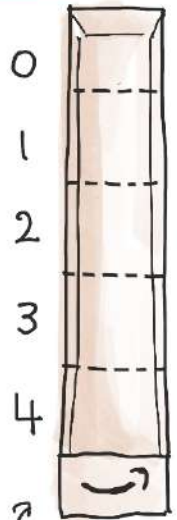
① Let's create an array of size 5.

We're going to add 🐱 data.

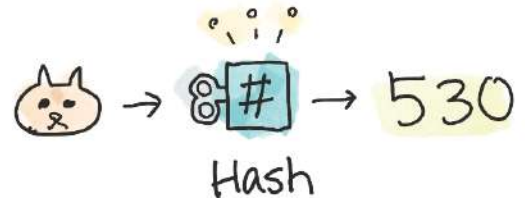
key = "Tabby"

value = "pizza"

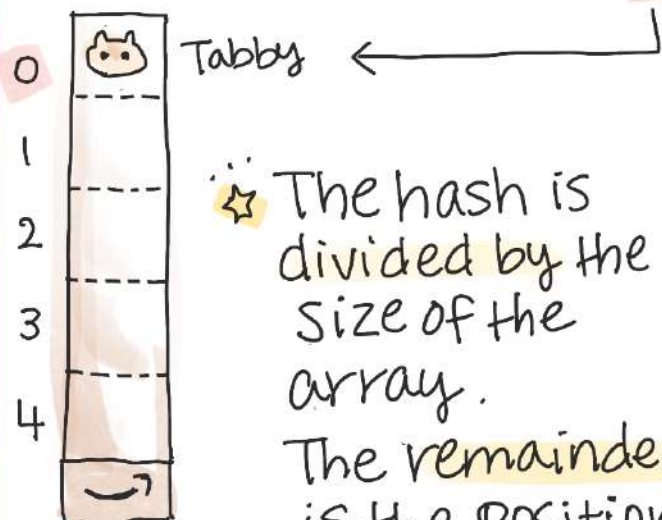
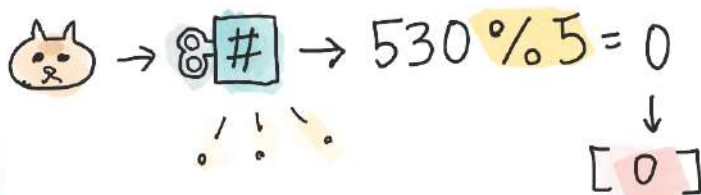
some data
Let's say, favorite food!



② Calculate the hash value by using the key, "Tabby".
e.g. ASCII code, MD5, SHA1

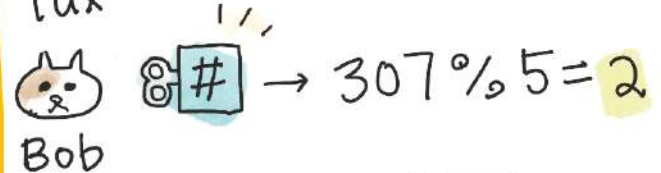
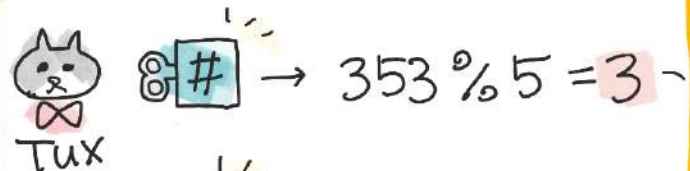


③ Use modulo to pick a position in the array!

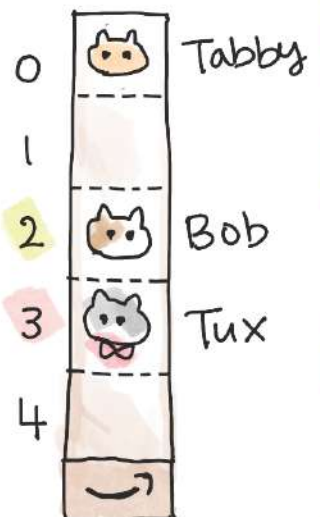


☆ The hash is divided by the size of the array.
The remainder is the position!

④ Let's add more data.



Use the same method to add more 🐱



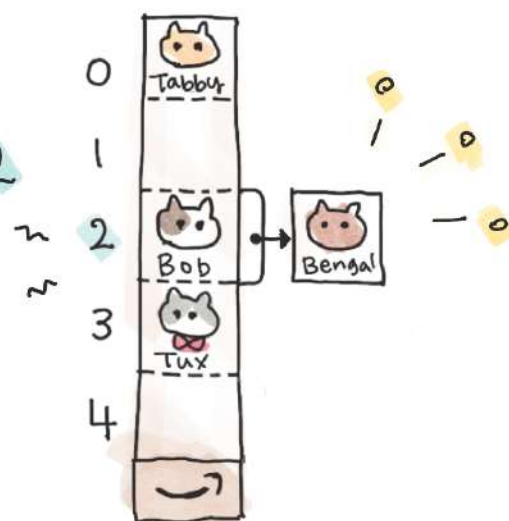
★ Collision!

Hash Table

Now we want to add more data.
Let's add "Bengal".

🐱 "Bengal" → $617\%5 = 2$

But [2] slot has been taken
by "Bob" already! = collision!
so let's chain Bengal next
to Bob! = chaining

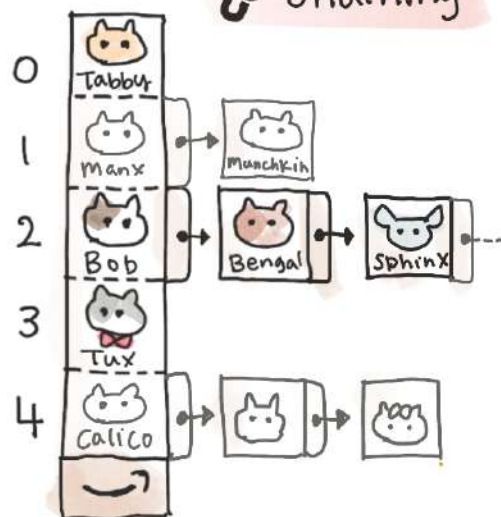


key: "Bengal"
Value: "Dosa"

"Sphinx"
"Fish +
Chips"

Keep
adding
data

chaining



🔍 Searching for data

★ Let's look up the value for "Bob"

① Get the hash → 307

② Get the index → $307\%5 = 2$

③ Look up Array [2] → found!

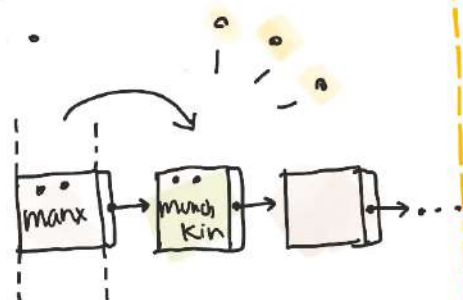
★ Let's look up "munchkin"

① Hash → 861

② Index → $861\%5 = 1$

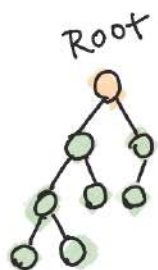
③ Array [1] → "Manx"

④ Operate a linear-search to find "munchkin"
↳ Average $O(n)$



Data Structures

Binary Heap



Binary tree

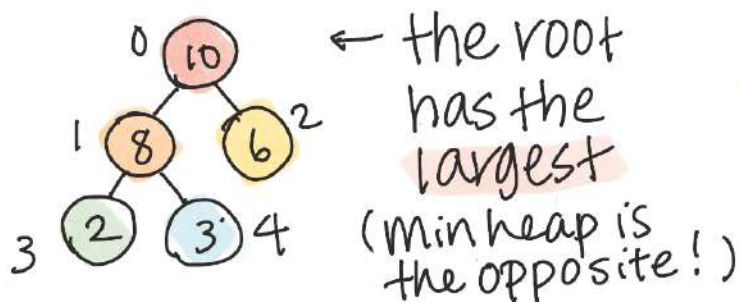
tree data structure
each node has at most 2 children

Binary search tree

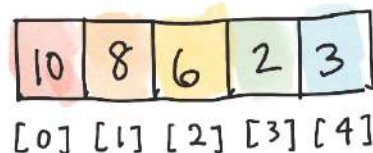
Binary heap

- ♥ Complete tree
- ♥ Min heap or max heap
- ♥ used for priority queue, heap sort etc.

★ Max heap

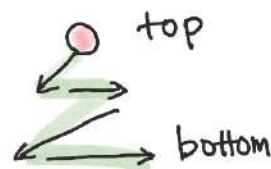


in array



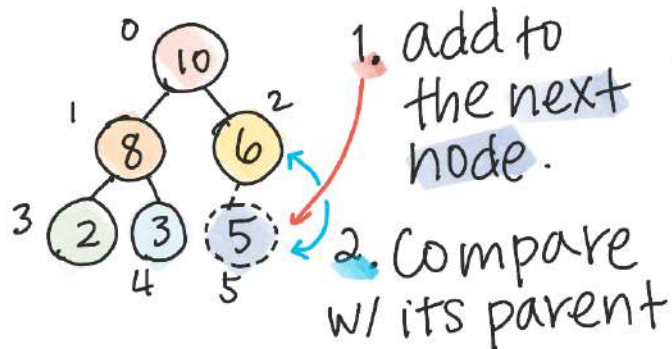
♥ each node has 0 - 2 children

♥ always fill top → bottom, left → right

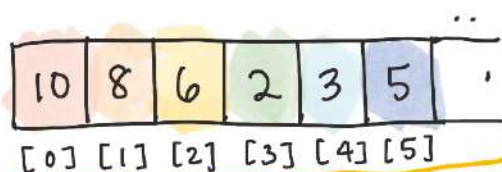


★ Insertion

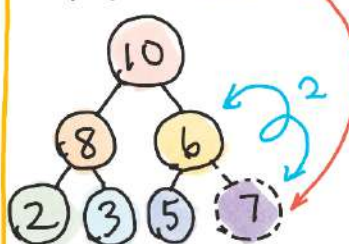
Let's add 5 to the heap!



1. add to the next node.
2. Compare w/ its parent
3. the parent is greater.
Cool, it's done!
Let's add more!



Add 7

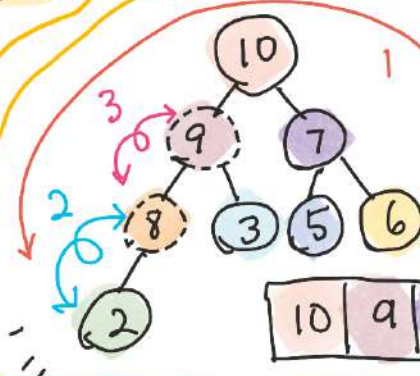
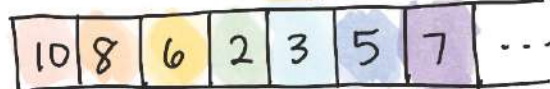


1. Add to the next node

2. Compare w/ parent.

Oh, no!

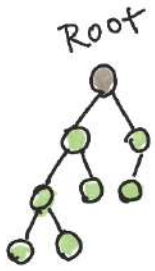
the parent is smaller than its child! Swap them !!!



Add to the next node & repeat the process!



Binary Search Tree



Binary tree

tree data structure
each node has at most 2 children

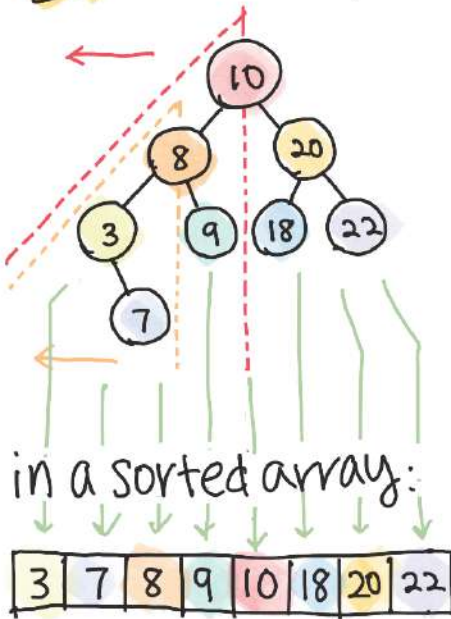
Binary heap

Binary Search Tree

a.k.a. ordered or sorted binary tree

fast look up
e.g. phone number lookup table by name

👍 Rule of thumb



★ each value of all nodes in the left subtrees is lesser

△ 10's left subtrees: 8, 3, 9, 7

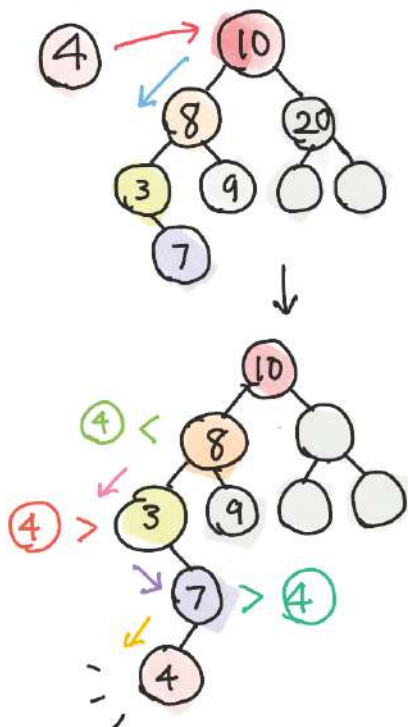
△ 8: 3, 7 ← smaller than parent

★ each value of all nodes in the right subtrees is larger

★ no duplicate values

★ Insertion → Always add to the lowest spot to be a leaf ~~🚫~~ No rearrange!

Let's add 4



1. Compare w/ the root first.

2. 4 < 10 so go left.

3. then compare w/ the next, 8

4. 4 < 8 so go left

5. Compare w/ the 3

6. 4 > 3 so go right.

7. Compare w/ the 7

8. 4 < 7, so add to the left! Done.

Complexity:

Ave. $O(\log n)$

Worst. $O(n)$

Refer link for more